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Chapter 6

The Continuity Struggle

6.12 Problem 6.12

Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Prove that $\lim_{x \rightarrow 0} f\left(\frac{1}{x}\right) = \lim_{x \rightarrow \infty} f(x)$. That is, prove that if one converges to some L , then the other also converges to L ; if one diverges to ∞ or to $-\infty$ or whose limit does not exist, then the other does the same.

6.12.1 Solution

Proof by Contradiction. Suppose for contradiction that the two limits are not equal.

$$\lim_{x \rightarrow 0^+} f\left(\frac{1}{x}\right) \neq \lim_{x \rightarrow \infty} f(x) \quad (6.1)$$

Suppose there exists a function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined as taking the reciprocal of its initial value.

$$g(x) := \frac{1}{x} \quad (6.2)$$

We can prove that $g(x)$ converges to 0 from the positive direction as x approaches ∞ . Suppose Let $\varepsilon > 0$ and let $\delta = \frac{1}{\varepsilon} > 0$. Then, for any x that satisfies $x > \delta$ and $x > 0$, we can take its reciprocal.

$$0 < \frac{1}{x} < \frac{1}{\delta} \quad (6.3)$$

Recall that, by the sequence squeeze theorem and the Archimedian Principle, for every sequence x_n that converges to ∞ , $\frac{1}{x_n}$ converges to 0, so we can replace $\frac{1}{\infty}$ with 0.

$$|g(x) - g(\infty)| = \left| \frac{1}{x} - \frac{1}{\infty} \right| \quad (6.4)$$

$$= \left| \frac{1}{x} - 0 \right| = \left| \frac{1}{x} \right| \quad (6.5)$$

This has an attached inequality from earlier.

$$|g(x) - g(\infty)| < \left| \frac{1}{\delta} \right| = \left| \frac{1}{\frac{1}{\varepsilon}} \right| = |\varepsilon| = \varepsilon \quad (6.6)$$

$$|g(x) - g(\infty)| < \varepsilon \quad (6.7)$$

Therefore, $g(x)$ converges to 0 from the positive direction as x approaches ∞ .

Suppose we made a sequence a_n that converges to 0 and another sequence b_n that converges to 0 from the positive direction as n approaches ∞ . $g(b_n)$ would in turn diverge to ∞ . We can rewrite our initial supposition.

$$\lim_{x \rightarrow 0^+} f\left(\frac{1}{x}\right) \neq \lim_{x \rightarrow \infty} f(x) \quad (6.8)$$

We can rewrite the latter term by substituting in $g(x)$.

$$\lim_{x \rightarrow 0^+} f\left(\frac{1}{x}\right) = \lim_{n \rightarrow \infty} \frac{1}{g(b_n)} = \lim_{n \rightarrow \infty} \frac{1}{\frac{1}{b_n}} = \lim_{n \rightarrow \infty} b_n \quad (6.9)$$

We can rewrite the former term by substitutring in a_n .

$$\lim_{x \rightarrow \infty} f(x) = \lim_{n \rightarrow \infty} a_n \quad (6.10)$$

Apply these to the inequality.

$$\lim_{n \rightarrow \infty} b_n \neq \lim_{n \rightarrow \infty} a_n \quad (6.11)$$

This contradicts the idea that functional limits are unique. Ergo, our initial assumption is false and $\lim_{x \rightarrow 0} f\left(\frac{1}{x}\right) = \lim_{x \rightarrow 1} f(x)$. OE Δ